Lecture 6: Error Propagation and Covariance

"Uncertainty"

Photograph by Michelle Lane

Physics 4730
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Propagation of Uncertainties

$$f = f(a, b, c \ldots)$$

$$S_f^2 = \left(\frac{\partial f}{\partial a}\right)^2 S_a^2 + \left(\frac{\partial f}{\partial b}\right)^2 S_b^2 + \left(\frac{\partial f}{\partial c}\right)^2 S_c^2 + \cdots$$

Where is this from?

- Consider function of one variable f(x)
- Assume errors small, so f(x) ~ linear

$$S_f = f(x + S_x) - f(x) \approx \frac{\Delta f}{\Delta x} \cdot S_x \to \frac{\partial f}{\partial x} \cdot S_x$$

Generally we express errors as positive, so...

$$S_f = \left| \frac{\partial f}{\partial x} \right| \cdot S_x$$

- Now, what about multivariate functions *e.g.* f = f(x,y)?
- Assume errors small, so f(x,y) ~ linear

$$f = f(x, y)$$

$$S_f = \left|\frac{\partial f}{\partial x}\right| \cdot S_x + \left|\frac{\partial f}{\partial y}\right| \cdot S_y$$

- e.g. Measure the heights x and y of two people
- What is the uncertainty in the sum of heights q=x+y?

$$q = x + y$$

$$S_q = \left|\frac{\partial q}{\partial x}\right| \cdot S_x + \left|\frac{\partial q}{\partial y}\right| \cdot S_y = S_x + S_y$$

$$q \overset{?}{=} (x + y) \pm (S_x + S_y)$$

Consider Two Cases:

$$q \overset{?}{=} (x+y) \pm (S_x + S_y)$$

I.

- x and y have correlated errors (*e.g.* ruler is off)
- Expression above gives upper bound to errors

II.

- x and y have uncorrelated errors
- Expression above overestimates error

How do we correctly treat functions of two (or more) variables?

Now, in the case where there are two independent variables and $q_i = q(x_i, y_i)$, we have

$$q_i \sim q(\overline{x}, \overline{y}) + \frac{\partial q}{\partial x}(x_i - \overline{x}) + \frac{\partial q}{\partial y}(y_i - \overline{y})$$

or, by noting that $\overline{q} = q(\overline{x}, \overline{y})$ (proof left as exercise), we get for the RMS of q

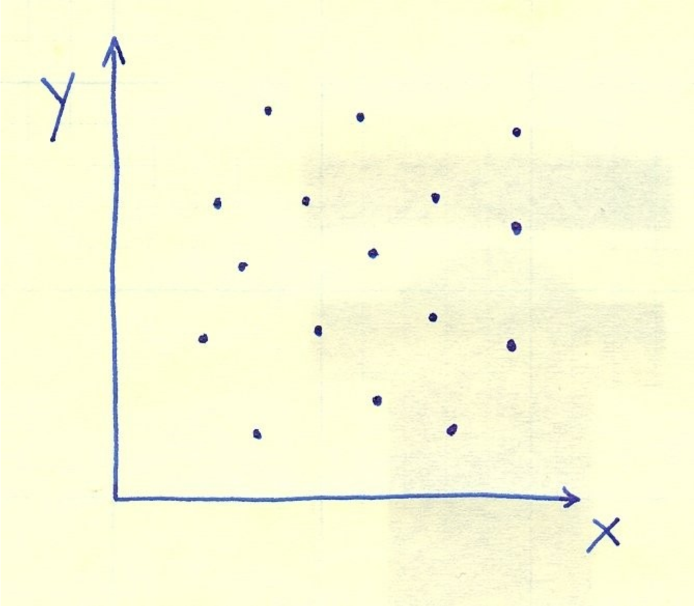
$$\begin{aligned}
\sigma_q^2 &= \frac{1}{N-1}\sum_{i=1}^{N}(q_i - \overline{q})^2 \\
&\sim \frac{1}{N-1}\sum_{i=1}^{N}\left[\frac{\partial q}{dx}(x_i - \overline{x}) + \frac{\partial q}{\partial y}(y_i - \overline{y})\right]^2 \\
&\sim \left(\frac{\partial q}{dx}\right)^2 \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2 + \left(\frac{\partial q}{dy}\right)^2 \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2 + 2\left(\frac{\partial q}{dx}\right)\left(\frac{\partial q}{dy}\right)\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y}) \\
&\sim \left(\frac{\partial q}{dx}\right)^2 \sigma_x^2 + \left(\frac{\partial q}{dy}\right)^2 \sigma_y^2 + 2\left(\frac{\partial q}{dx}\right)\left(\frac{\partial q}{dy}\right)\sigma_{xy}
\end{aligned}$$

where we have introduced a new term, the *covariance* of x and y:

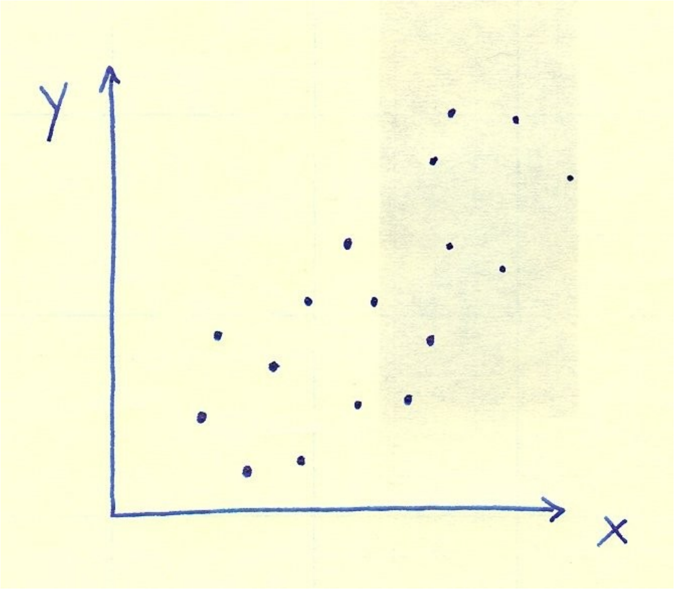
$$\sigma_{xy} \equiv \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y})$$

Covariance:

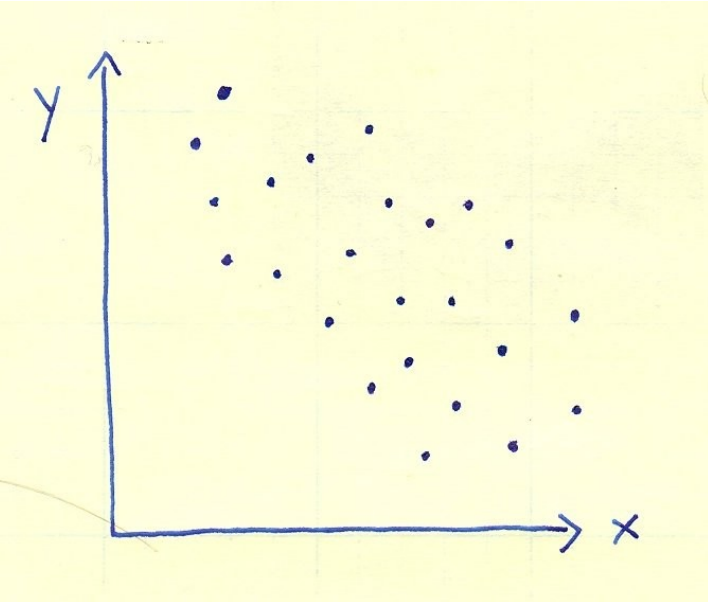
$$\sigma_{xy} \equiv \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y})$$



Zero Covariance



Positive Covariance



Negative Covariance

$$q = q(x, y)$$

"Old" Terms

New Term

$$\sigma_q^2 \sim \left(\frac{\partial q}{dx}\right)^2 \sigma_x^2 + \left(\frac{\partial q}{dy}\right)^2 \sigma_y^2 + 2\left(\frac{\partial q}{dx}\right)\left(\frac{\partial q}{dy}\right)\sigma_{xy}$$

$$\sigma_x^2 \equiv \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2$$

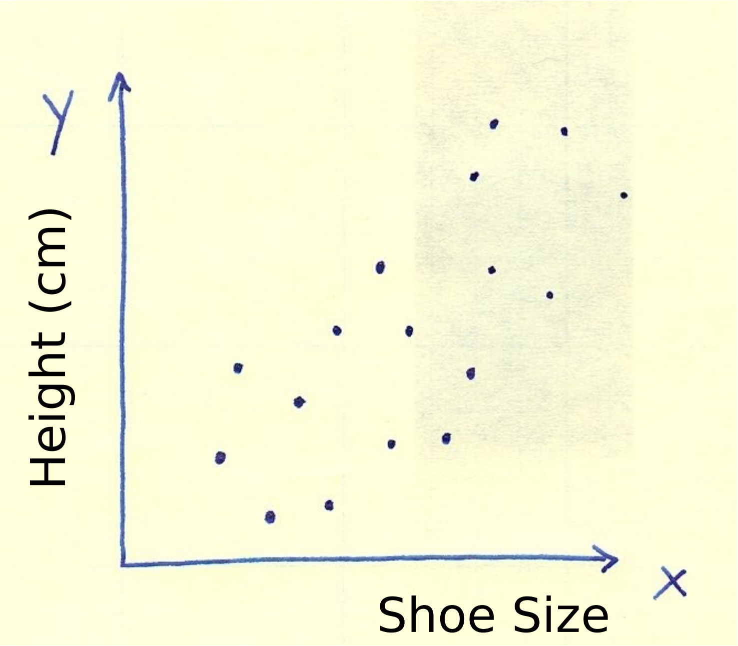
$$\sigma_y^2 \equiv \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2$$

$$\sigma_{xy} \equiv \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y})$$

"Covariance"
- form intuitive
- tends to vanish if x & y vary independently
- reduces to former result if no correlation

Quantifying Correlation in terms of Covariance

- We want to know if two variables (not uncertainties!) are likely to be linearly related.
- e.g. Shoe size and height



- Introduce "coefficient of linear correlation" (Pearson's coefficient) *r*:

$$r = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum (x_i - \overline{x})^2} \sqrt{\sum (y_i - \overline{y})^2}}$$

$$-1 \leq r \leq 1$$

perfect anticorrelation

perfect correlation

Significance of r Depends on N_{points}

Chance probability or “p-value”

Table C. The percentage probability $Prob_N(|r| \geq r_o)$ that N measurements of two uncorrelated variables give a correlation coefficient with $|r| \geq r_o$, as a function of N and r_o. (Blanks indicate probabilities less than 0.05%.)

N \ r_o	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
3	100	94	87	81	74	67	59	51	41	29	0
4	100	90	80	70	60	50	40	30	20	10	0
5	100	87	75	62	50	39	28	19	10	3.7	0
6	100	85	70	56	43	31	21	12	5.6	1.4	0
7	100	83	67	51	37	25	15	8.0	3.1	0.6	0
8	100	81	63	47	33	21	12	5.3	1.7	0.2	0
9	100	80	61	43	29	17	8.8	3.6	1.0	0.1	0
10	100	78	58	40	25	14	6.7	2.4	0.5		0
11	100	77	56	37	22	12	5.1	1.6	0.3		0
12	100	76	53	34	20	9.8	3.9	1.1	0.2		0
13	100	75	51	32	18	8.2	3.0	0.8	0.1		0
14	100	73	49	30	16	6.9	2.3	0.5	0.1		0
15	100	72	47	28	14	5.8	1.8	0.4			0
16	100	71	46	26	12	4.9	1.4	0.3			0
17	100	70	44	24	11	4.1	1.1	0.2			0
18	100	69	43	23	10	3.5	0.8	0.1			0
19	100	68	41	21	9.0	2.9	0.7	0.1			0
20	100	67	40	20	8.1	2.5	0.5	0.1			0
25	100	63	34	15	4.8	1.1	0.2				0
30	100	60	29	11	2.9	0.5					0
35	100	57	25	8.0	1.7	0.2					0
40	100	54	22	6.0	1.1	0.1					0
45	100	51	19	4.5	0.6						0
	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	
50	100	73	49	30	16	8.0	3.4	1.3	0.4	0.1	
60	100	70	45	25	13	5.4	2.0	0.6	0.2		
70	100	68	41	22	9.7	3.7	1.2	0.3	0.1		
80	100	66	38	18	7.5	2.5	0.7	0.1			
90	100	64	35	16	5.9	1.7	0.4	0.1			
100	100	62	32	14	4.6	1.2	0.2				

Table from Taylor, *Intro to Error Analysis*

Some Python Tools

For the arrays x, y:

• numpy.average(x)	output is scalar
• numpy.cov(x,y)	2x2 matrix
• scipy.stats.pearsonr(x,y)	2-vector

Today's Lab

A professor wants to know if telling a lot of jokes in class gets him good student evaluations. Over several years, he records the number of jokes he tells and compares it with the overall performance rating which he is given by the students. His data are collected in the table below:

Year	1	2	3	4	5
Number of Jokes	12	18	33	27	39
Rating	5.1	4.1	3.8	3.3	3.6

Quantitatively evaluate the correlation between jokes and class evaluations. Should the instructor keep telling jokes, or doesn't it matter?